

INTAPAN INJECTION 1ML (10MG/ML) AMPOULE

DESCRIPTION: Clear, colourless liquid.

COMPOSITION:

Each ampoule contains Nalbuphine Hydrochloride 10 mg per ml.

PHARMACODYNAMICS:

Nalbuphine HCl has the effect of lowering the cardiac work load and can be used immediately in myocardial infarction (use caution where emesis is involved). Haemodynamic studies in patients with severe arteriosclerotic heart changes reveal that Nalbuphine has circulatory effects similar to those of morphine, i.e. a minimal decrease in oxygen consumption, cardiac index, left ventricular end diastolic pressure and cardiac work. The antagonist activity of Nalbuphine is $\frac{1}{4}$ as potent as nalorphine and approximately $\frac{1}{40}$ that of naloxone.

PHARMACOKINETICS:

Nalbuphine HCl is a potent analgesic. Its analgesic potency is essentially equivalent to that of morphine on a milligram basis. The onset of action occurs within 2-3 min. after administration and in < 15 min. following SC or IM injection. The plasma half-life of Nalbuphine is $\frac{1}{4}$ as potent as nalorphine and 10 times that of pentazocine.

INDICATION:

Moderate to severe pain. Preoperative and postoperative analgesics, as a supplement to surgical anaesthesia, for obstetrical analgesia, during labour.

RECOMMENDED DOSAGE:

The usual recommended adult dose is 10 mg for 70 kg individual, administered SC, IM or IV; this dose may be repeated every 3-6 hours as necessary. Dosage should be adjusted according to the severity of the pain, physical status of the patients, and other medications which the patients may be receiving, see Interactions (other CNS depressants). In non-tolerant individuals, the recommended single maximum dose is 20 mg, with a maximum total daily dose of 160 mg.

The use of Nalbuphine as a supplement to balanced anaesthesia requires larger doses than those recommended for analgesic. Induction doses of nalbuphine range from 0.3 – 3 mg/kg IV to be administered over 10 to 15 min period with maintenance doses of 0.25 – 0.50 mg/kg in single IV administration as required. The use of Nalbuphine may be followed by respiratory depression which can be reversed with the narcotic antagonist Narcan (Naloxone HCl).

Patients dependent on narcotics: Patients who have been taking narcotics chronically may experience withdrawal symptoms upon the administration of Nalbuphine.

If unduly troublesome, narcotic withdrawal symptoms can be controlled by the slow IV administration of small increments of morphine, until relief occurs. If the previous analgesic was morphine, meperidine, codeine, or other narcotic with similar duration of activity, $\frac{1}{4}$ of the anticipated dose of nalbuphine can be administered initially and the patient observed for signs of withdrawal, i.e. abdominal cramps, nausea and vomiting, lacrimation, rhinorrhoea, anxiety, restlessness, elevation of temperature or piloerection. If untoward symptoms do not occur, progressively larger doses may be tried at appropriate intervals until the desired level of analgesia is obtained with nalbuphine.

Central Nervous System effects: Nervousness, depression, restlessness, crying, euphoria, floating, hostility, unusual dreams, confusion, faintness, hallucinations, dysphoria, feeling of heaviness, numbness, tingling, unreality. The incidence of psychotomimetic effects, e.g. unreality, depersonalization, delusions, dysphoria and hallucinations has been shown to be less than that which occurs with pentazocine. It is not recommended do not perform machinery work or drive after taking this medicine.

ROUTE OF ADMINISTRATION:

For s.c./i.m./i.v. injection

CONTRAINDICATIONS:

Hypersensitivity to Nalbuphine.

WARNINGS & PRECAUTIONS:

Nalbuphine should be administered as a supplement to general anaesthesia only by persons specifically trained in the use of IV anaesthetic and management of the respiratory effects of potent opioids.

Drug dependence: Nalbuphine has been shown to have a low-abuse potential when compared with drugs which are not mixed agonist-antagonist. It has been reports that Nalbuphine's potential for abuse would be less than that of codeine and propoxyphene. Psychological and physical dependence and tolerance may follow the abuse or misuse of nalbuphine. Therefore, caution should be observed in prescribing it for emotionally unstable patients, or for individuals with a history of narcotic abuse. Such patients should be closely supervised when long term therapy is contemplated. Care should be taken to avoid increases in dosage or frequency of administration which in susceptible individuals may result in physical dependence. Abrupt discontinuation of nalbuphine following prolonged use, has been followed by symptoms of narcotic withdrawal, i.e. abdominal cramps, nausea and vomiting, rhinorrhoea, lacrimation, restlessness, anxiety, elevated temperature and piloerection.

Use in emergency procedures: Maintain patient under observation until recovered from nalbuphine effects that would affect driving or other potentially dangerous tasks.

Use in ambulatory patients: Nalbuphine may impair the mental or physical abilities required for the performance of potentially dangerous tasks, e.g. driving a car or operating machinery. Therefore, it should be administered with caution to ambulatory patients who should be warned to avoid such hazards.

Head injury and increased intracranial pressure: The possible respiratory depressant effects and the potential of potent analgesics to elevate cerebrospinal fluid pressure (resulting from vasodilation following CO₂ retention) may be markedly exaggerated in the presence of head injury, intracranial lesions or a pre-existing increase in intracranial pressure. Furthermore, potent analgesics can produce effects which may obscure the clinical course of patients with head injuries. Therefore nalbuphine should be used in these circumstances only when essential, and then should be administered with extreme caution.

Precautions:

Impaired respiration: 10 mg / 70 kg of nalbuphine causes some respiratory depression approximately equivalent to usual doses of morphine. However, in contrast to morphine, respiratory depression is not appreciably increased with higher doses of nalbuphine. Respiratory depression induced by nalbuphine can be reversed by Naloxone HCl when indicated. Nalbuphine should be administered with caution at low doses to patients with impaired respiration (e.g. from other medication, uremia, bronchial asthma, severe infection, cyanosis or bronchial respiratory obstructions).

Impaired renal or hepatic function: Because nalbuphine is metabolized in the liver and excreted by the kidneys, patients with renal or liver dysfunction may over-react to customary doses. Therefore, in these individuals, nalbuphine should be used with caution and administered in reduced amounts.

Myocardial infarction: As with all potent analgesics, nalbuphine should be used with caution in patients with myocardial infarction who have nausea or vomiting.

Biliary tract surgery: As with all narcotic analgesics, nalbuphine should be used with caution in patients about to undergo surgery of the biliary tract since it may cause spasm of the sphincter of Oddi.

Cardiovascular system: During elevation of nalbuphine in analgesia, a higher incidence of bradycardia has been reported in patients who did not receive atropine preoperatively or in the postoperative period.

Use in pregnancy: Safe use of nalbuphine in pregnancy (other than labour) has not been established. Although animal reproductive studies have not revealed teratogenic or embryotoxic effects, nalbuphine should only be administered to pregnant women (other than in labour) when, in the judgement of the physician, the potential benefits outweigh the potential hazards.

Use during labour of delivery: Nalbuphine can produce respiratory depression in the neonate. It should be used with caution in women delivering premature infants.

Use in children: Sufficient clinical experience to support administration of nalbuphine in children is not available at present.

Risk from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of INTAPAN INJECTION with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserved concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosage and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risk of respiratory depression and sedation when INTAPAN INJECTION is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepines have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of INTAPAN INJECTION with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See interactions with other medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue INTAPAN INJECTION if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid and allow adrenal function to recover and continued corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function / Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

INTERACTION WITH OTHER MEDICAMENTS:

Although nalbuphine possesses narcotic antagonist activity, there is evidence that in nondependent patients it will not antagonize a narcotic analgesic administered just before, concurrently, or just after injection of nalbuphine. Therefore, patients receiving a narcotic analgesic, general anaesthetics, phenothiazines, or other tranquilizers, sedatives, hypnotics or other CNS depressants (including alcohol) concomitantly with nalbuphine may exhibit an additive effect. When such combined therapy is contemplated, the dose of one or both agents should be reduced.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepine and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue INTAPAN if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue (See Warnings and Precautions)).

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Safe use of Nalbuphine in pregnancy (other than labour) has not been established. Although animal reproductive studies have not revealed teratogenic or embryotoxic effects, nalbuphine should only be administered to pregnant women (other than in labour) when, in the judgement of the physician, the potential benefits outweigh the potential hazards.

Use during labour of delivery: Nalbuphine can produce respiratory depression in neonate. It should be used with caution in women delivering premature infants.

SIDE EFFECTS

The most frequent adverse reaction in treated with nalbuphine is sedation. Less frequent reactions are: sweaty / clammy, nausea / vomiting, dizziness / vertigo, dry mouth and headache.

Other adverse reactions which may occur are:

CNS effects: Nervousness, depression, restlessness, crying, euphoria, floating, hostility, unusual dreams, confusion, faintness, hallucinations, dysphoria, feeling of heaviness, numbness, tingling, unreality. The incidence of psychotomimetic effects, e.g. unreality, depersonalization, delusions, dysphoria and hallucinations has been shown to be less than that which occurs with pentazocine.

Cardiovascular: Hypertension, hypotension, bradycardia, tachycardia, pulmonary oedema.

Gastrointestinal: Cramps, dyspepsia, bitter taste

Respiratory: Depression, dyspnoea, asthma

Dermatological: Itching, burning, urticaria

Miscellaneous: Speech difficulty, urinary urgency, blurred vision, flushing and warmth.

Post-marketing Experience:

Serotonin syndrome (See Warnings and Precautions)

Adrenal insufficiency (See Warning and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

INCOMPATIBILITIES:

Unknown

SYMPTOMS AND TREATMENT OF OVERDOSE:

The administration of single SC doses of 72 mg of nalbuphine to 8 normal subjects has been reported to have resulted primarily in symptoms of sleepiness and mild dysphoria. The immediate IV administration of Naloxone HCl is a specific antidote. Oxygen, IV fluids, vasopressors and other supportive measures should be used as indicated.

STORAGE CONDITIONS:

Store below 30°C. Protect from excessive light.

Keep out of reach of children. Jauhkan daripada kanak-kanak.

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

1ml amp in box of 10's

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang,

Selangor, MALAYSIA.